

Lesson One

Nature of Software Engineering Management

**Effective Project Management:
Traditional, Agile, Extreme**

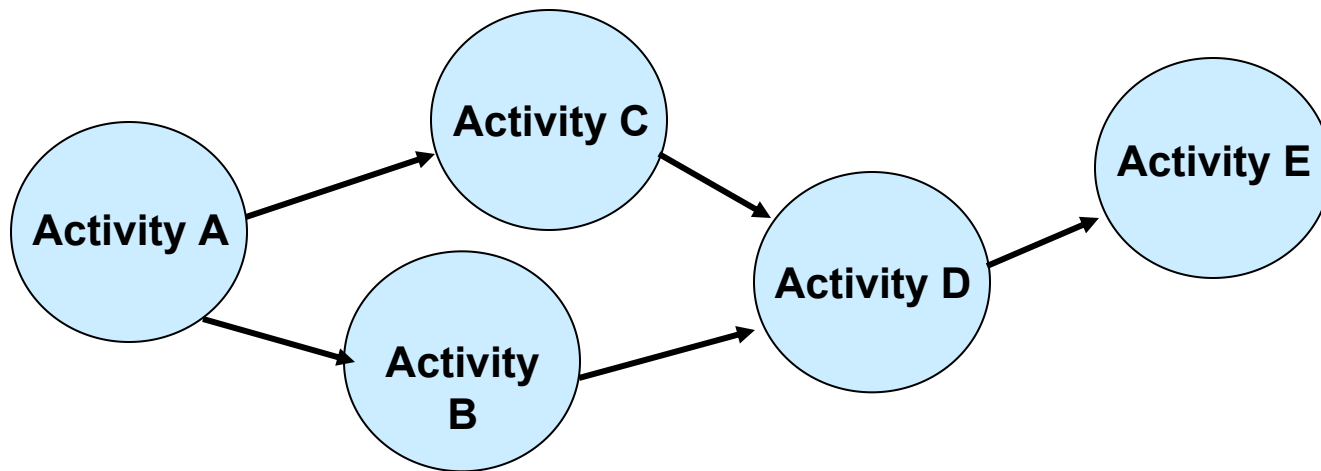
Ch01: What Is a Project?

***Presented by
Bowen DU***

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Defining a Project

A project is a sequence of unique, complex, and connected activities having one goal or purpose and that must be completed by a specific time, within budget, and according to specification.



What's missing from this definition?

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A Business-focused Definition of a Project

A project is a sequence of finite dependent activities whose successful completion results in the delivery of the **expected business value** that validated doing the project.

The major focus of every project is the satisfaction of client needs through the generation of the expected business value. That will be the primary metric for assessing project performance and progress.

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A Business-focused Definition of a Project

Class Exercise #2

Which are projects in the following cases?

1. Producing an edition of a newspaper.
2. Getting married / Planning a wedding.
3. Designing and implementing a computer system.
4. Hosting a holiday party.
5. Designing and producing a brochure.
6. Executing an environmental clean-up of a contaminated(污染) site.
7. Holding a high school reunion.
8. Performing a series of surgeries on an accident victim.

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An Intuitive(直观的) View of the Project Landscape

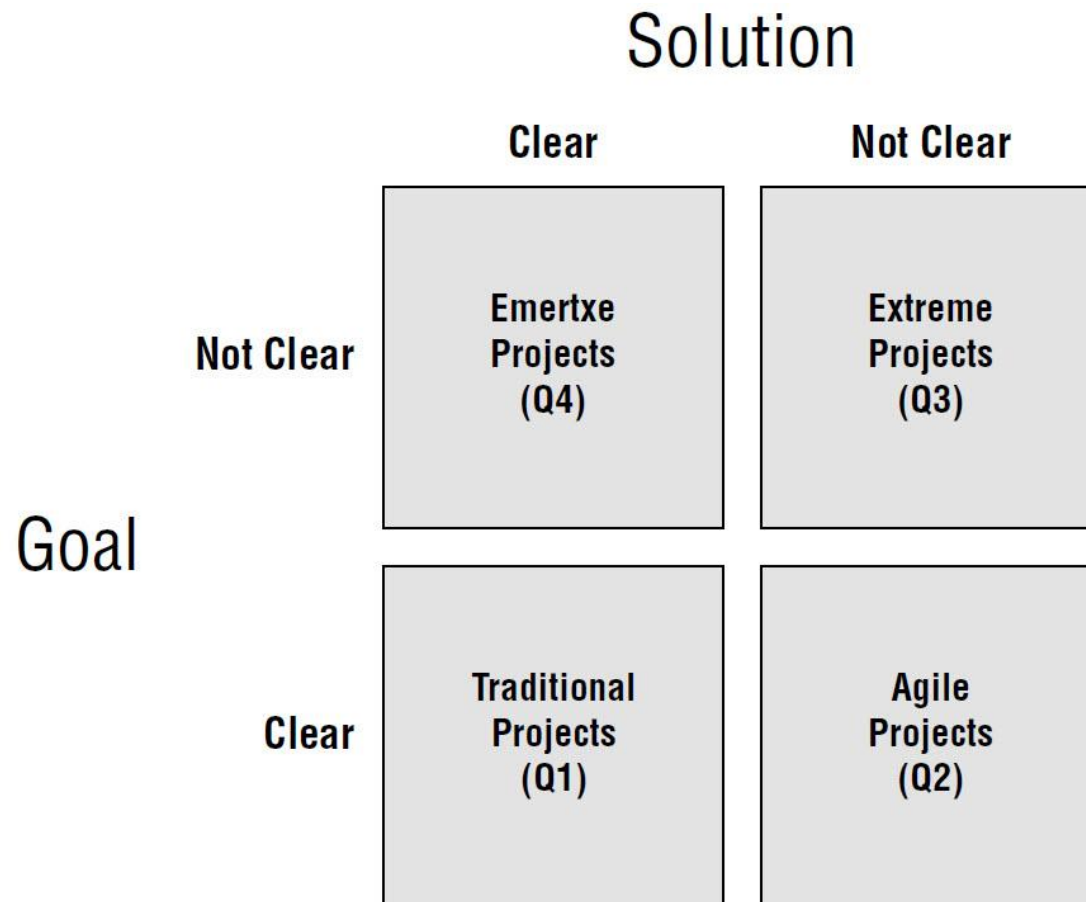
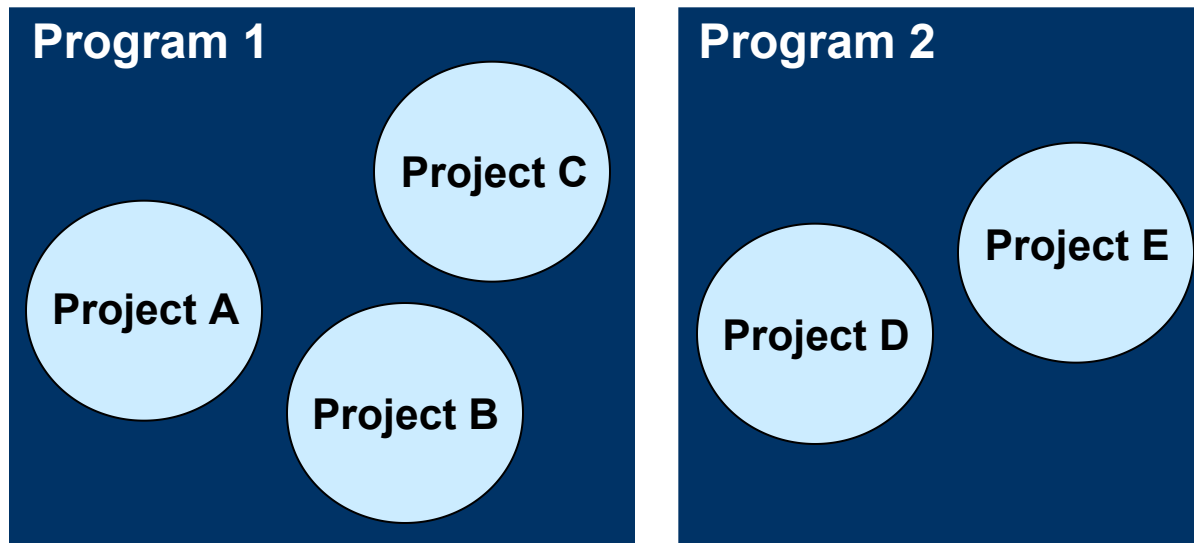


Figure 1.1: The four quadrants of the project landscape

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Defining a Program

A program is a collection of related projects that share a common goal or purpose.



Defining a Portfolio

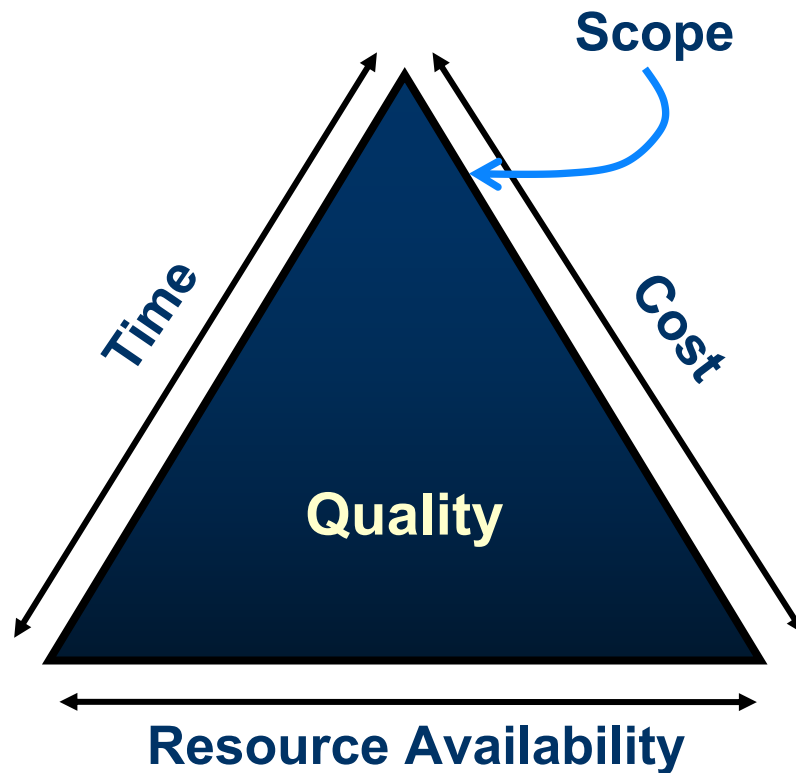
A portfolio is a collection of projects that share some common link to one another.

For example:

- Same business unit
- New product development projects
- R & D projects
- Maintenance projects
- Process improvement projects
- Staffed from the same resource pool
- Same budget

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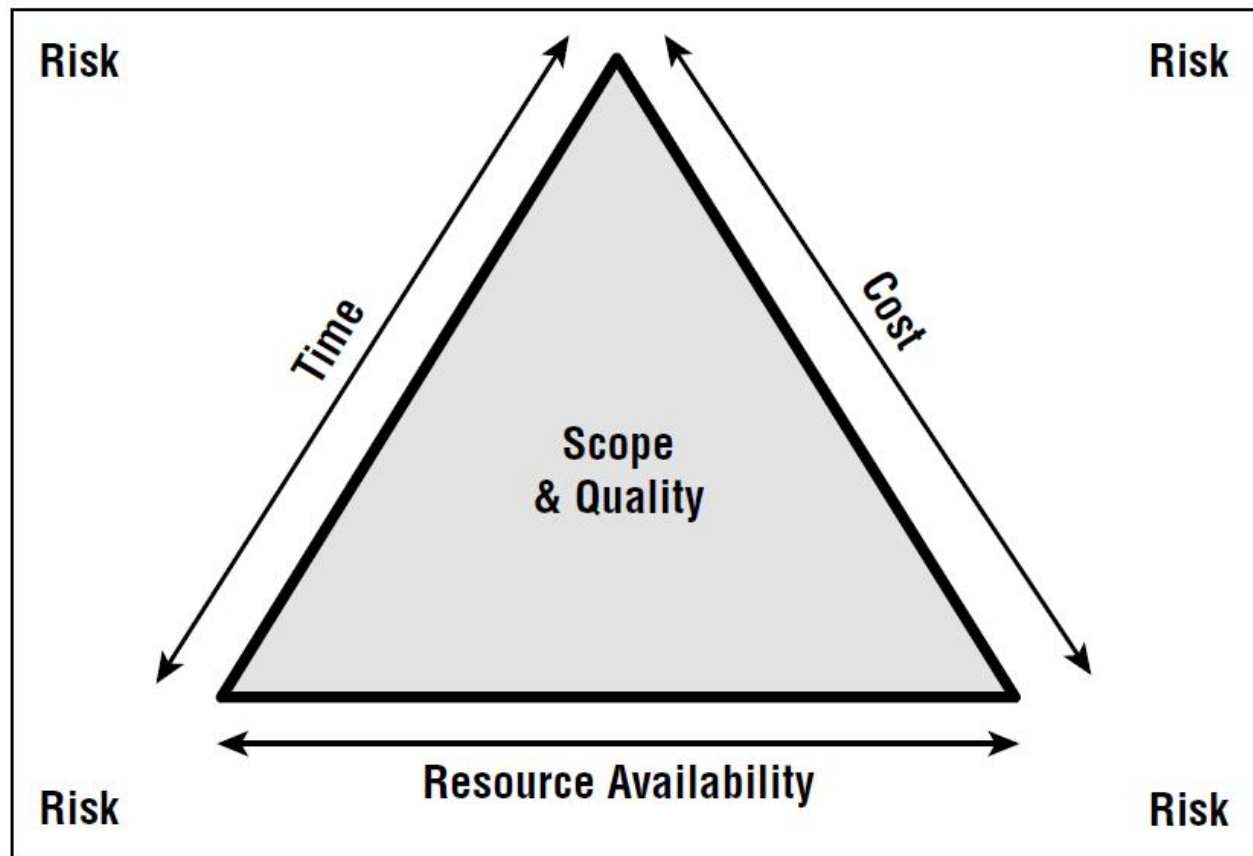
The Scope Triangle is a System in Balance



- The Scope Triangle is a system in balance.
- The lengths of the three sides exactly bound scope and quality.
- Change in the variables will cause the system to be out of balance.

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Envisioning the Scope Triangle is a System in Balance



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Prioritizing the Scope Triangle

- The scope triangle can be used:
 - To build a problem resolution strategy
 - For scope change impact analysis

Variable \ Priority	Critical (1)	(2)	(3)	(4)	Flexible (5)
Scope				X	
Quality			X		
Time	X				
Cost					X
Resource Availability		X			

Figure 1.3: Prioritized Scope Triangle variables

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Project Classification

- To adopt a “one size fits all(一刀切)” approach to every project is just asking for trouble.
- Your approach to managing any project must adapt to the characteristics of the project.
- A classification rule can help you choose that suitable approach.



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Classification rules by Project Characteristics

- Risk
- Business Value
- Duration
- Complexity
- Technology used
- Number of departments affected
- Cost

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Example Project Classes and Definitions

Table 1.1: Example of Project Classes and Definitions

CLASS	DURATION	RISK	COMPLEXITY	TECHNOLOGY	LIKELIHOOD OF PROBLEMS
Type A	> 18 months	High	High	Breakthrough	Certain
Type B	9–18 months	Medium	Medium	Current	Likely
Type C	3–9 months	Low	Low	Best of breed	Some
Type D	< 3 months	Very low	Very low	Practical	Few

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Required and Optional Processes

Project Management Process	Project Classification			
	IV	III	II	I
Define				
Conditions of Satisfaction	R	R	O	O
Project Overview Statement	R	R	R	R
Approval of Request	R	R	R	R
Plan				
Conduct Planning Session	R	R	O	O
Prepare Project Proposal	R	R	R	R
Approval of Proposal	R	R	R	R
Launch				
Kick-Off Meeting	R	R	O	O
Task Schedule	R	R	R	R
Resource Assignments	R	R	R	O
Statements of Work	R	O	O	O
Monitor/Control				
Status Reporting	R	R	R	R
Project Team Meetings	R	R	O	O
Approval of Deliverables	R	R	R	R
Close				
Post-implementation Audit	R	R	R	R
Project Notebook	R	R	O	O

R = Required O = Optional

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Classification by Project Application

- Installing software
- Recruiting and hiring
- Setting up a hardware system in a field office
- Soliciting, evaluating, and selecting vendors
- Updating a corporate procedure
- Developing application systems

The Contemporary Project Environment

- High Speed
- High Change
- Lower Cost
- Increasing Levels of Complexity
- More Uncertainty

Summary of Chapter 1

- Defining a project
- An intuitive view of the project landscape
- Defining a program
- Defining a portfolio
- Understanding the scope triangle
- The importance of classifying projects
- The contemporary project environment

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Assignments of Chapter 1

Paper reading

Project start-up Getting going effectively. Peter W G Morris, 1984

You can download this paper from Canvas”课外阅读文献”, We will discuss this paper in next lesson.

Team building

Before 23:59 on **March 16th**, submit the list of team members for the course project and selected topic on canvas by replying the assignment.